

Kanu Madhok

952-303-1045 • madhok.kanu@gmail.com • github.com/kmadhok • linkedin.com/in/kanu-madhok

PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Built rigorous evaluation infrastructure for LLM systems: a 23-case golden-query evaluation suite plus human-in-the-loop validation for a production analytics agent over 67 BigQuery tables (sub-agent orchestration, 8 deterministic SQL rules) — the regression-testing discipline that keeps a probabilistic system measurable and reproducible.
- Built and shipped an autonomous Jira agent with agentic RAG: 6-gate triage, MCP-driven context gathering, hybrid FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution; closed 400+ requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10. Designed for reproducibility with decision logging and tracing.
- Built a registry-governed semantic layer that boxes the LLM into strict-JSON intent (never SQL) with deterministic validation and a human review gate, making the single probabilistic step regression-testable — applying experimental rigor to a production system.
- Built a production headless KPI monitor (6-hour cron) with root-cause analysis, Slack alerts, and BigQuery-persisted decision logs for end-to-end observability; designed a 3-tier event-driven v2 self-configuring via a registry.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Designed and ran a persona-driven multi-LLM (OpenAI + Gemini) experiment with 500+ participants: controlled A/B tests measuring persuasion, with rigorous experimental design and analysis — controlled experimentation that persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Built a RAG-to-proposal generator deployed on AWS EC2; owned design and evaluation end-to-end; cut proposal time ~30%; **Best in Show — UChicago MSADS Capstone.**

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); feature engineering, model validation; retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Research & experimentation: controlled A/B experimentation, multi-LLM evaluation, golden-set / regression evaluation harnesses, human-in-the-loop validation, experimental design & analysis

LLM systems: RAG & vector search (embeddings + FAISS + BM25 + RRF), agent orchestration & tool use, MCP server design, prompt engineering, multi-model routing, context engineering

Engineering: Python, FastAPI, REST APIs, Cloud Run, GCP / AWS, CI/CD (pytest), decision logging / tracing / observability, version control

Data: BigQuery, dbt, SQL, PySpark, Airflow, Databricks, MySQL

LLMs / dev: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI; Claude Code, Cursor, LangChain

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL